

1. Show the electric field, represented as ψ in the equation below, of (a) a plane wave and (b) a Hertzian dipole satisfy the Sommerfeld Radiation Condition,

$$\lim_{R \rightarrow \infty} R^{\frac{n-1}{2}} \left(\frac{\partial \psi}{\partial R} + jk\psi \right) = 0 \quad n=1, 2, \text{ or } 3 \text{ dimensions}$$

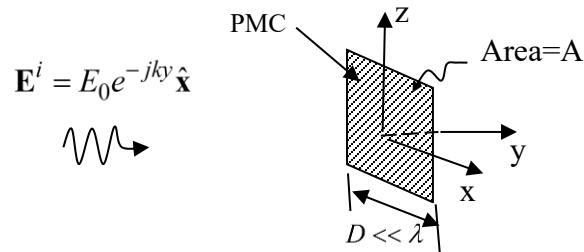
where

$$R = x, y, \text{ or } z \text{ when } n=1$$

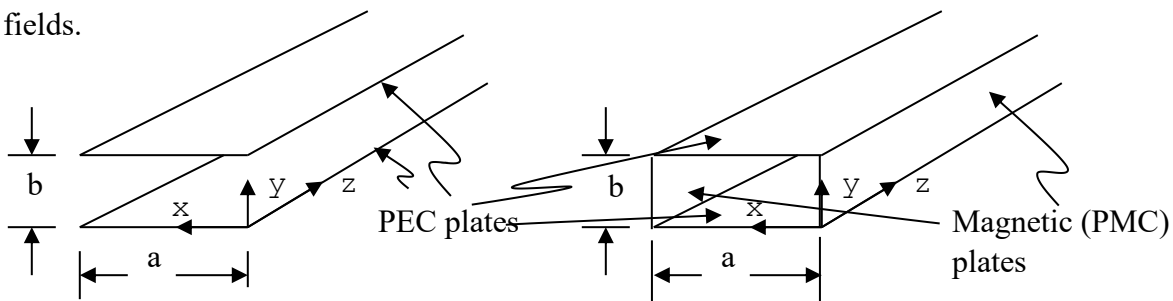
$$R = \rho \text{ when } n=2$$

$$R = r \text{ when } n=3$$

2. Solve the physical optics problem on page 9-1 in the notes, assuming the plate is a perfect magnetic conductor (PMC) and use the incident field $\mathbf{E}^i = E_0 e^{-jk_y \hat{\mathbf{y}}}$.



3. A parallel plate waveguide is first order approximation of a microstrip line. Use the method presented in class to find the (a) TM mode fields (b) TE mode fields in an air filled parallel-plate waveguide shown in the figure below using the approximation that in a parallel-plate waveguide the fields are constant in the x -direction, so $\frac{\partial}{\partial x} = 0$ and therefore $k_x = 0$ for all fields.



The approximation that the fields are not a function of x when the sides of the waveguide are open allows the assumption that the side walls can be approximated as perfect magnetic conductors.

4. Solve for the fields in a rectangular waveguide in the TM case. Go through the complete waveguide method as was shown in the notes for the TE case. (a) Write down the pilot function and the equations for the electric and magnetic fields in terms of the potentials $\mathbf{A}$ or $\mathbf{F}$. (b) Below the equations for $\mathbf{E}$ and $\mathbf{H}$ write the field expression for each component of $\mathbf{E}$ and $\mathbf{H}$ in terms of the pilot function ψ . (c) Write down the differential equation for ψ , separate variables, write

down the dispersion relation, the three separated equations and ψ in terms of the separated functions. (d) Write ψ in terms of the sine and cosine solutions for the two transverse separate solutions (e) Write down the boundary conditions you are going to use (f) apply each boundary condition to find the final function ψ multiplied by the exponential solution for the separated equation function of z . (g) Put ψ into the field components, take the derivatives and give the final expressions for the fields.

